

Xiyuan Bao

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[Personal Website](#)

Department of Earth and Planetary Sciences, Harvard University

EDUCATION	Ph.D. in Geophysics , University of California, Los Angeles	2018-Mar 2024
	M.S. in Geophysics , University of California, Los Angeles	2018-2020
	B.S. in Geophysics with Honors , University of Science and Technology of China	2014-2018
	Minor in Computer Science , University of Science and Technology of China	2014-2018
EXPERIENCE	Gordon and Betty Moore Foundation Postdoctoral Fellow , Harvard University	Oct 2026-
	Reginald A. Daly Postdoctoral Fellow , Harvard University	Oct 2024-Sep 2026
	Postdoctoral Scholar , University of California, Los Angeles	Mar 2024-Sep 2024
	Graduate Research Assistant , University of California, Los Angeles	Sep 2018-Mar 2024
	Undergrad Research Assistant , University of Science and Technology of China	Sep 2017-May 2018
	Visiting Undergraduate , University of Illinois, Urbana-Champaign	Jun-Aug 2017
RESEARCH INTERESTS	• Dynamics of the Solid Earth Interior • Surface Processes and Coupling with Planetary Interiors • Experimental, Computational and Machine Learning Methods in Geophysics	
GRANTS, HONORS & AWARDS	• Harvard Data Science Initiative-Amazon Web Service (HDSI-AWS) Impact Computing Project (Lead Proposal Writer, PI ineligible due to postdoctoral status, \$209,302) 2026-2028 • Gordon and Betty Moore Foundation Postdoctoral Fellowship (\$192,308) 2026-2028 • Reginald A. Daly Postdoctoral Research Fellowship, Harvard University (\$171,500) 2024-2026 • Study of the Earth's Deep Interior Section Award for Graduate Research, AGU 2023 • Computers and Geosciences Research Scholarship, IAMG 2022 • Jiuzhang Zhao Talent Program in Earth and Space Sciences, USTC 2015-2018 • Award from National Undergraduate Innovation Training Program, USTC 2017-2018 • Earth Science Climbing Scholarship, USTC 2015, 2017 • Silver Scholarship for Outstanding Students, USTC 2016 • First Prize of Physics Innovation Research Experiment Thesis Competition, USTC 2016	
PUBLICATIONS	* denotes publications for which I served as the (co-)corresponding author; † denotes students or postdocs (co-)advised by me. [9] Bao, X.*, Wamba, M. D. & Stracke, A. A joint geophysical-geochemical deep-mantle zoning map beneath East Africa and the Indian Ocean. <i>Science</i> 393, 684–689. https://www.science.org/doi/full/10.1126/science.aea5466 (2026). [8] Coulson, S., Lloyd, A., Bao, X., Mitrovica, J. X., Dangendorf, S., Pan, L., Valencic, N., Tamisiea, M. E., Al-Attar, D. & Heathcotte, D. Inverting Sea Surface Height Data Yields Greenland Ice Mass Changes (1993-2019): A Proof of Concept. <i>Geophysical Journal International</i>. https://academic.oup.com/gji/advance-article/doi/10.1093/gji/ggag187/8677305 (2026). [7] Lucas, E., Richards, F., Cederberg, G., Bao, X., Hoggard, M. J., Tsuji, S. R. J., Latychev, K., Tsuji, L. J. S. & Mitrovica, J. Emergence of Antarctic mineral resources in a warming world. <i>Nature Climate Change</i>. https://www.nature.com/articles/s41558-026-02569-1 (2026). [6] Bao, X.* & Lithgow-Bertelloni, C. R. Self-correction of the optical distortion effect of thermal plumes in particle image velocimetry. <i>Physics of Fluids</i>. https://doi.org/10.1063/5.0233759 (2024).	

- [5] **Bao, X.***. Giant impact and Earth's mysterious blobs: An interdisciplinary revelation. *Science Bulletin* **69**, 293–294. <https://cdn.sciengine.com/doi/10.1016/j.scib.2023.12.028> (2024).
- [4] **Bao, X.***, Mittal, T. & Lithgow-Bertelloni, C. R. Determining Mid-Ocean Ridge Geography from Upper Mantle Temperature. *Earth and Planetary Science Letters* **641**, 118823. <https://www.sciencedirect.com/science/article/pii/S0012821X24002565> (2024).
- [3] Zhang, X., Brown, E. L., Zhang, J., Lin, J., **Bao, X.** & Sager, W. W. Magmatism of Shatsky Rise controlled by plume-ridge interactions. *Nature Geoscience*. <https://www.nature.com/articles/s41561-023-01286-0> (2023).
- [2] Li, S., Li, J., Ferrand, T. P., Zhou, T., Lv, M., Xi, Z., Maguire, R., Han, G., Li, J., **Bao, X.**, *et al.* Deep geophysical anomalies beneath the Changbaishan Volcano. *Journal of Geophysical Research: Solid Earth*, e2022JB025671. <https://agupubs.onlinelibrary.wiley.com/doi/full/10.1029/2022JB025671> (2023).
- [1] **Bao, X.***, Lithgow-Bertelloni, C. R., Jackson, M. G. & Romanowicz, B. On the relative temperatures of Earth's volcanic hotspots and mid-ocean ridges. *Science* **375**, 57–61. ISSN: 0036-8075. <https://www.science.org/doi/10.1126/science.abj8944> (2022).

MANUSCRIPTS
SUBMITTED

- [7] Wurmser, S. S., Al Asad, M., **Bao, X.**, Creveling, J., Mitrovica, J. X. & Tryon, C. A. *Reopening the Southern Gate: Early Pleistocene Sea Level and Hominin Dispersal Across the Red Sea*
- [6] Acquadro, J.[†], **Bao, X.*** & Deng, J. *Machine-Learned Local Environments Reveal Framework Controls on MgSiO₃ Liquid-Crystal Similarity Under review at PNAS NEXUS.*
- [5] Powell, R., Cederberg, G., Dangendorf, S., Lucas, E., **Bao, X.**, Creveling, J., Tsuji, S., Tsuji, L. & Mitrovica, J. X. *Disproportionate sea level rise in small island developing states of the Pacific and Indian Oceans driven by Greenland and Antarctic ice mass loss Under review at Communications Earth & Environment.*
- [4] Cai, R., **Bao, X.**, Foley, S. F., Giuliani, A., Jiang, S., Pearson, D. G. & Liu, J. *Evidence for a highly fusible fundamental constituent in the convective mantle. In Revision.*
- [3] **Bao, X.*** & Lithgow-Bertelloni, C. R. *A method for measuring and analyzing four-dimensional flow fields viscous in Rayleigh-Bénard Convection Revision Under Review.*
- [2] Wang, W., Zhang, X., Xu, F., Cheng, Z., **Bao, X.**, Stracke, A., Zha, C., Zhan, F. & Lin, J. *Ultra-depleted mantle beneath the Marion Rise and its implications for ocean rise uplift*
- [1] **Bao, X.*** & Lithgow-Bertelloni, C. R. *Morphology, interactions, and evolution of laminar thermal plumes revealed by four-dimensional velocity measurements and analysis: Application to Earth's mantle Revision Under Review.*

MANUSCRIPTS
IN PREP

- [2] **Bao, X.***, Coulson, S., Valencic, N., Mousavi, M., Dangendorf, S., Lloyd, A. & Mitrovica, J. *Neural Operators Reveal Scale-Dependent Limits on Recovering Greenland Ice Loss from Sea-Level Fingerprints*
- [1] **Bao, X.***, Cao, Z., Gourley, K. C., Gutierrez, S., Liang, Y., Zhou, D., Buffett, B. A., Mittal, T., Li, J., Pierrehumbert, R., Schaefer, L., Orman, J. V. & Foley, B. J. *Coupled Evolution of Earth's Hydrogen Distribution and Thermal History.*

ORAL
PRESENTATIONS

- [19] **Invited:** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Geodynamics Seminar (Lamont-Doherty Earth Observatory, Palisades, NY, US, Feb. 2026).
- [18] **Invited:** *What Powers Earth's Hotspot Volcanism? What Does It Reveal About the Deep Mantle?* Geo. Sci. Special Seminar (University of Chicago, Chicago, IL, US, Feb. 2026).
- [17] *Inverting Sea Level Fingerprints for Ice Mass Changes Using Neural Operators: How Much Can We Recover and How Unique Is It?* AGU Fall Meeting 2025 (New Orleans, LA, US, Dec. 2025).
- [16] **Invited:** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Brownbag Seminar (Princeton University, Princeton, NJ, US, Nov. 2025).
- [15] **Invited:** *Deep Provenance of East African and Indian Ocean Hotspots through Tomography, Graph Theory, and Geochemical Cross-validation* Geophysics Hour (Arizona State University, Tempe, AZ, US, Nov. 2025).
- [14] **Invited:** *Plumes over LLSVPs: Supercharging multi-scale flow resolution with hybrid experimental-adjoint digital twins* Geophysics Seminar (Brown University, RI, US).
- [13] *Plumes over LLSVPs: Supercharging multi-scale flow resolution with hybrid experimental-adjoint digital twins* BiSEPPS Seminar (Harvard University, MA, US, Nov. 2024).
- [12] **Invited:** *A Song of Ice and Fire: Operator Learning for Viscous Flow from Mantle Convection to Ice Dynamics* Daly Special Seminar (Harvard University, MA, US, Feb. 2024).
- [11] **Invited:** *Hotspots from Top to Bottom* Geophysics Seminar (University of Science and Technology of China, Anhui, China, Jan. 2024).
- [10] **Invited:** *Illuminating Mantle Convection: Unraveling Plume Dynamics over LLSVPs in the Laboratory* AGU Fall Meeting 2023 (San Francisco, CA, US, Dec. 2023).
- [9] **Invited:** *Hotspots from Top to Bottom* Earth Sciences Seminar (University of Southern California, CA, US, Oct. 2023).
- [8] *The plume zoo of thermal LLSVPs* AGU Fall Meeting 2022 (Chicago, IL, US, Dec. 2022), V26A–02.
- [7] *Coupled Evolution of Earth's Hydrogen Distribution and Thermal History* AGU Fall Meeting 2022 (Chicago, IL, US, Dec. 2022), MR33A–06.
- [6] **Invited:** *Hotspots from Top to Bottom* Seismo Lab Seminar (California Institute of Technology, CA, US, Nov. 2022).
- [5] **Invited:** *Are hotspots hotter than ridges?* IGCP 648 Virtual Seminar Series (Curtin University, Australia, Nov. 2022), Series 5, Seminar 5. <https://www.youtube.com/watch?v=30B21c7NGIY>.
- [4] *Experimental Investigation of Purely Thermal LLSVPs* UCLA EPSS Seismology and Tectonics Seminar (UCLA, CA, US, May 2022).
- [3] *Accurate Prediction of Ocean Basins Using Upper Mantle Potential Temperatures* AGU Fall Meeting 2021 (New Orleans, LA, US, Dec. 2021), DI34A–05.
- [2] *Accurate Prediction of Ocean Basins Using Upper Mantle Potential Temperatures* UCLA EPSS Geology/Geophysics Seminar (UCLA, CA, US, Oct. 2021).
- [1] *Correlation of geochemical signals with MORB and OIB temperatures* UCLA EPSS Geocheminar (UCLA, CA, US, Oct. 2019).

POSTER

- PRESENTATIONS [13] **Bao, X.**, Wamba, M. D. & Stracke, A. in. AGU Fall Meeting 2025 (New Orleans, LA, US, Dec. 2025).
- [12] **Bao, X.**, Coulson, S., Valencic, N., Mitrovica, J., Dangendorf, S., Mousavi, M. & Lloyd, A. in. 2025 Gordon Research Seminar: Interior of the Earth (June 2025).
- [11] **Bao, X.** & Lithgow-Bertelloni, C. R. in. AGU Fall Meeting 2024 (Washington DC, US, Dec. 2024).
- [10] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2024 Ada Lovelace Workshop on Modelling Mantle and Lithosphere Dynamics (University of California, Berkeley, CA, US, Sept. 2024).
- [9] **Bao, X.** in. AGU Fall Meeting 2023 (San Francisco, CA, US, Dec. 2023).
- [8] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2023 Gordon Research Seminar: Interior of the Earth (Mount Holyoke College, MA, US, June 2023).
- [7] **Bao, X.** & Lithgow-Bertelloni, C. R. in. Seismo Lab Centennial (California Institute of Technology, US, Nov. 2022).
- [6] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2022 Ada Lovelace Workshop on Numerical Modelling of Mantle and Lithosphere Dynamics (Hévíz, Hungary, Aug. 2022).
- [5] **Bao, X.** & Lithgow-Bertelloni, C. R. in. CIDER2022 (University of California, Berkeley, CA, US, July 2022).
- [4] **Bao, X.**, Lithgow-Bertelloni, C. R. & Jackson, M. G. in. AGU Fall Meeting 2020 (San Francisco, CA, US, Dec. 2020), DI007–0006.
- [3] Wang, Y., Liu, L. & **Bao, X.** in. AGU Fall Meeting 2020 (San Francisco, CA, US, Dec. 2020), DI005–0005.
- [2] **Bao, X.**, Lithgow-Bertelloni, C. R. & Jackson, M. G. in. AGU Fall Meeting 2019 (San Francisco, CA, US, Dec. 2019), DI41D–0028.
- [1] **Bao, X.** & Lithgow-Bertelloni, C. R. in. 2019 Gordon Research Seminar: Interior of the Earth (Mount Holyoke College, MA, US, June 2019).

PROFESSIONAL SERVICE

- **Guest Editor**, *PNAS Nexus*, First Call for Papers, in *Machine Learning and Geosciences*
- **Primary Session Convener and Chair**, *Advances in Machine Learning for Solid Earth Geoscience*, AGU Fall Meeting 2025
- **Primary Session Convener and Chair**, *Advances in Deep Earth–Surface Interactions*, AGU Fall Meeting 2024
- **Discussion Leader**, *The Structure and Composition of the Lower Mantle*, Gordon Research Seminar 2023
- **Primary Session Convener and Chair**, *Advances in Mantle Convection and Planetary Evolution*, AGU Fall Meeting 2022
- **Reviewer**, *Nature Geoscience*; *Science Advances*; *Journal of Geophysical Research: Solid Earth*; *Geophysical Journal International*; *Planetary Science Journal*; *Physics of Fluids*; *Earth and Planetary Physics*; *Frontiers in Earth Science*
- **Reviewer**, National Science Foundation, Geophysics Program

- OTHER SERVICE • Committee Member of UCLA EPSS Family Mentorship Program (EFMP)

OUTREACH	• Media Interview about our <i>AI-based LLSVP map from Harvard EPS</i>	Sep 2026
	• <i>Rheology, Plate Tectonics with Food</i> Saturday Science Academy, UCLA/Charles R. Drew University	Mar 2023
	• <i>Why Hawaii is a volcano? How do Alaskan volcanos differ?</i> EYU, UCLA	Nov 2022
	• Media Interview about our <i>cold hotspots</i> , e.g. <i>ScienceNews</i> , <i>New Scientist</i> ^{1,2} <i>Popular Science</i> , and <i>Deutschlandfunk</i>	Jan 2022
	• Invited Outreach Article <i>Rocks beneath volcanic hotspots can be surprisingly cool</i>	Jan 2022
	• <i>The sweet smell of Earth's mantle: Why is Hawaii a volcano?</i> EYU, UCLA	Nov 2020, 2021
	• Lab visit from Tokyo Tech. UCLA	Feb 2020
	• <i>Why is Hawaii a volcano?</i> Explore Your Universe, UCLA	Nov 2019
TEACHING EXPERIENCE	• Blue Planet: Introduction to Oceanography , Teaching Assistant, UCLA	Spring 2020
	• Earthquakes , Teaching Assistant, UCLA	Spring 2019, Fall 2021
	• AI for Earth and Planetary Sciences , Guest Lecturer, Harvard University	Spring 2026
MENTORING EXPERIENCE	• Natasha Valencic, PhD student at Harvard, <i>Separating Decadal Ice-Melt Fingerprints from Stereodynamic Sea-Level Change with Machine Learning</i>	June 2026 -
	• Doyoon Park, PhD student at Princeton, <i>Viscosity of Davemaoite Under Lower Mantle Conditions from Surrogate Machine Learning Molecular Dynamics</i>	Nov 2025 -
	• Jake Acquadro, PhD student at Princeton, <i>Machine-Learned Local Environments Reveal Framework Controls on MgSiO₃ Liquid-Crystal Similarity</i>	Nov 2025 -
	• Kavya Agarwal, visiting undergrad at UCLA, <i>Compositional and thermodynamical effect on mantle plumes; Critical plume merge distance with shadowgraph</i> ; now PhD student at Yale	Sep 2021 - Aug 2022
	• Wan Ki (Arthur) Lo, undergrad at UCLA, <i>Dynamics of Adjacent Mantle Plumes through Physical and Computer Simulations</i> , won 2021 Harold and Mayla Sullwold Scholarship; now PhD student at PSU	Jan 2020 - Jun 2022
	• Jade Wight, undergrad at UCLA, <i>Modeling Mantle Plumes and Their Effect on Dynamic Topography</i> , won 2020 Deane Oberste-Lehn Scholarship; past PhD student at UHawaii and now geologist at AE-COM	Oct 2019 - Jun 2021